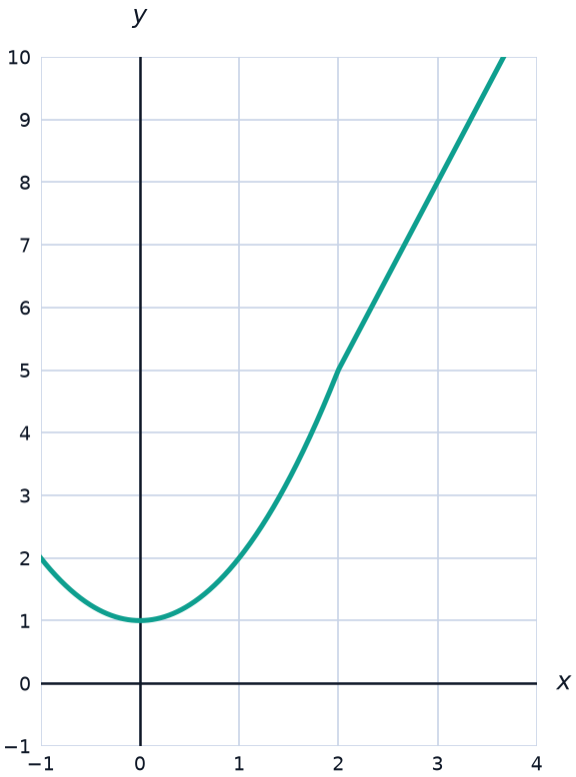
Continuity

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Testing continuity**

Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.



a Find $\lim_{x \to 2^-} f(x)$.

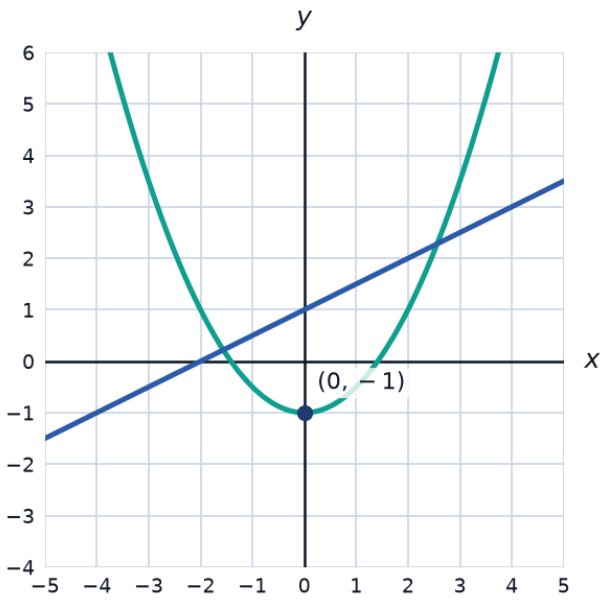
b Find $\lim_{x \to 2^+} f(x)$.

c Is f continuous at $x = 2$? Justify.

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2 [Foundational] • Target: Testing continuity

The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k.



3 [Foundational] • Target: Types of discontinuity and the Intermediate Value Theorem

Classify the discontinuity of each function at the given point:

a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$

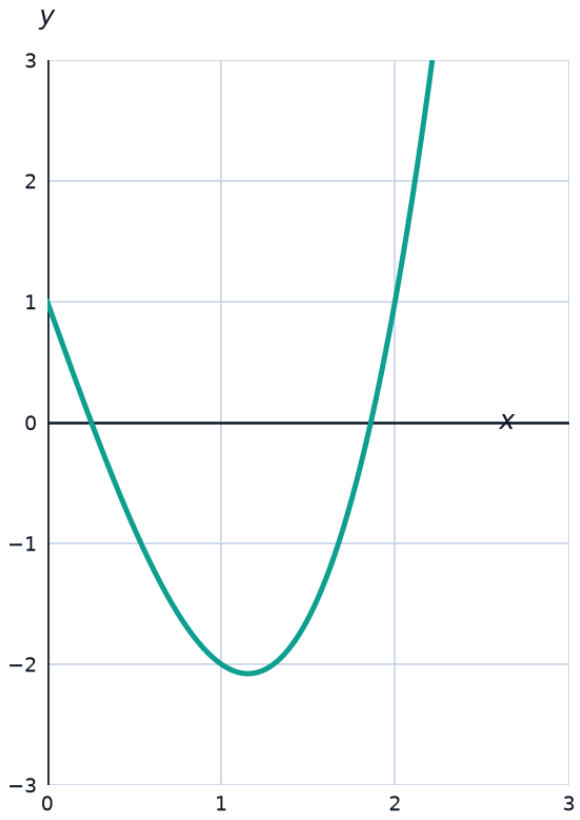
b $f(x) = \frac{1}{x - 2}$ at $x = 2$

c $f(x) = \lfloor x \rfloor$ at $x = 3$

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4 [Foundational] • Target: Types of discontinuity and the Intermediate Value Theorem

Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.



5 [Foundational] • Target: Continuity on an interval

State the largest interval(s) on which each function is continuous:

a $f(x) = \frac{1}{x-4}$

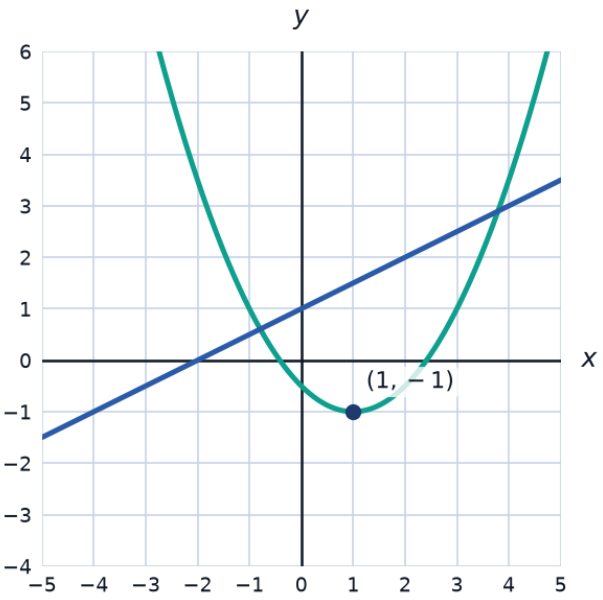
b $f(x) = \sqrt{x-3}$

c $f(x) = \ln(x+2)$

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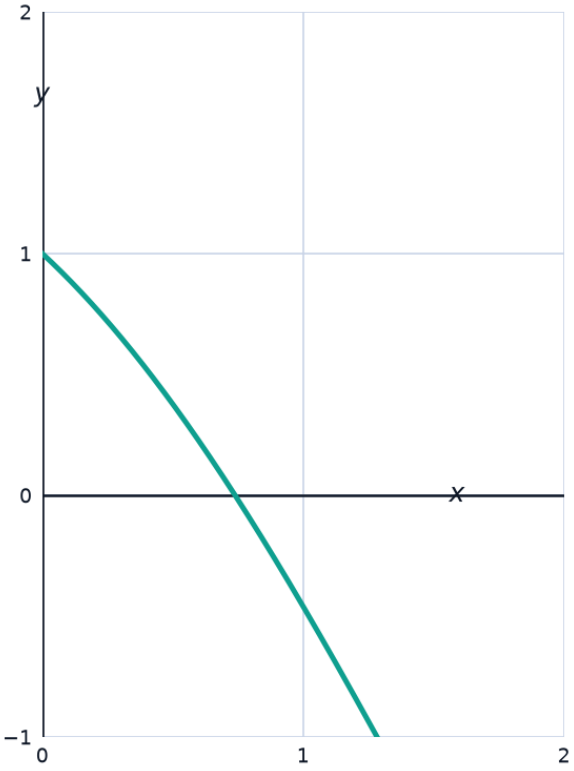
6 [Foundational] • Target: Continuity on an interval

Find the value of c that makes $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$, and $f(3) = c$, continuous at $x = 3$.



7 [Foundational] • Target: Applying the Intermediate Value Theorem

Show that $f(x) = \cos x - x$ has a root in the interval $(0, 1)$.



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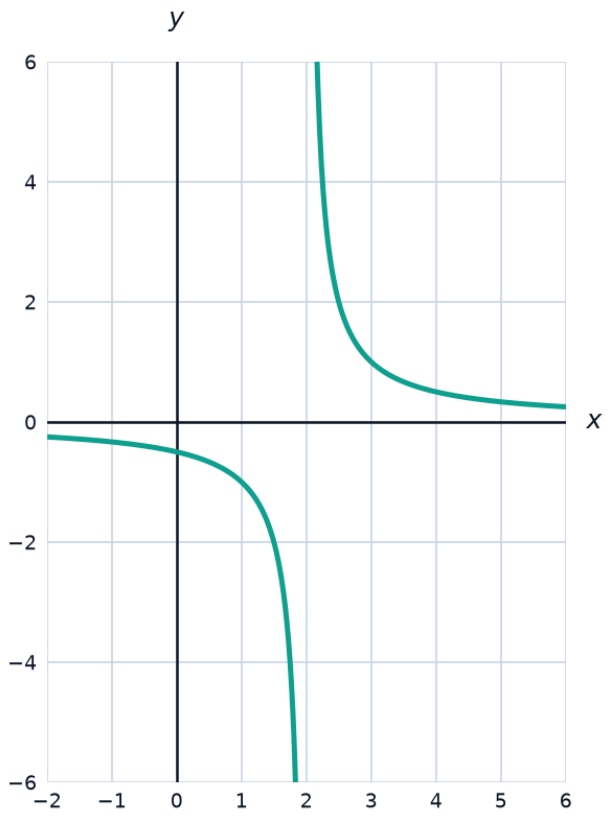
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Applying the Intermediate Value Theorem**

A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.

9 **[Intermediate]** • **Target: Visualizing discontinuities**

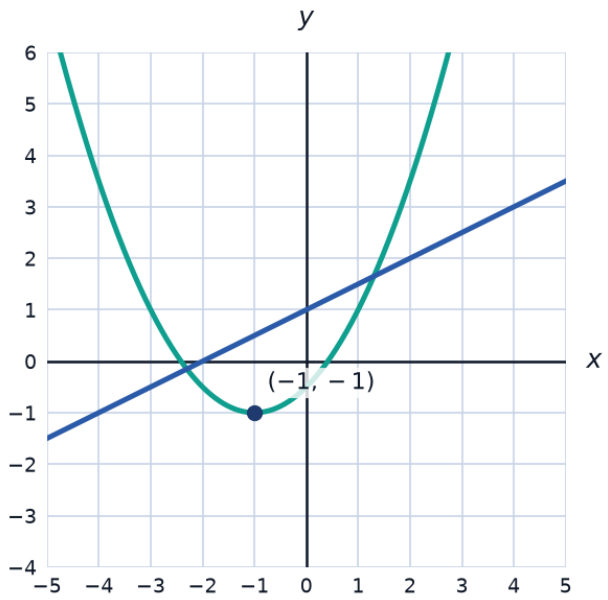
The function $f(x) = \frac{1}{x-2}$ (with an infinite discontinuity at $x = 2$) is graphed below. Use the graph to describe the one-sided behavior near $x = 2$.



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10 [Intermediate] • Target: Continuity of composite and combined functions

Explain why the composition of two continuous functions is continuous, and use this to argue that $h(x) = \sqrt{x^2 + 1}$ is continuous everywhere.



11 [Intermediate] • Target: Removable discontinuities and redefinition

The function $f(x) = \frac{x^2 + x - 6}{x - 2}$ has a removable discontinuity at $x = 2$. Find the value that, if assigned to $f(2)$, would make f continuous there.

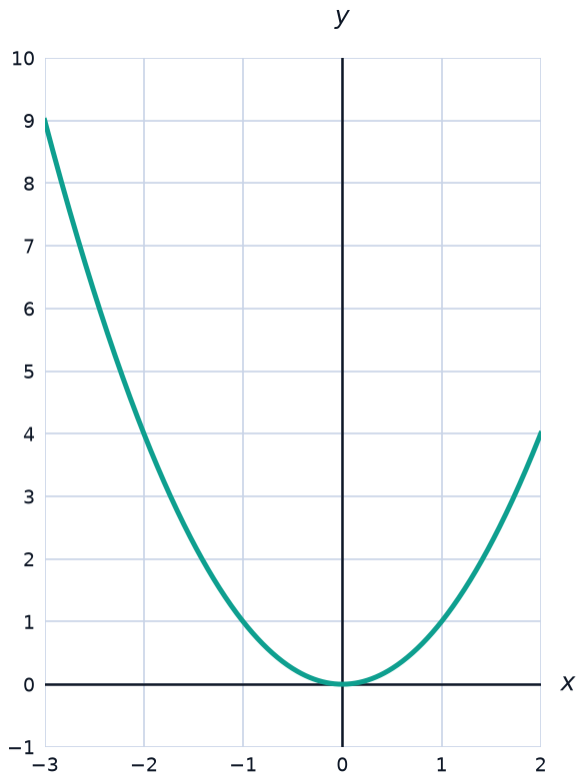
12 [Intermediate] • Target: Removable discontinuities and redefinition

Find all values of x where $f(x) = \frac{x^2 - 16}{x^2 - 2x - 8}$ has a discontinuity, and classify each as removable or non-removable.

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13 [Intermediate] • Target: The Extreme Value Theorem (a continuity consequence)

State the Extreme Value Theorem, and explain why $f(x) = x^2$ is guaranteed to have an absolute maximum and minimum on $[-3, 2]$.



14 [Intermediate] • Target: Types of discontinuity and the Intermediate Value Theorem

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{3^{-n}(x-3)^n}{n}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

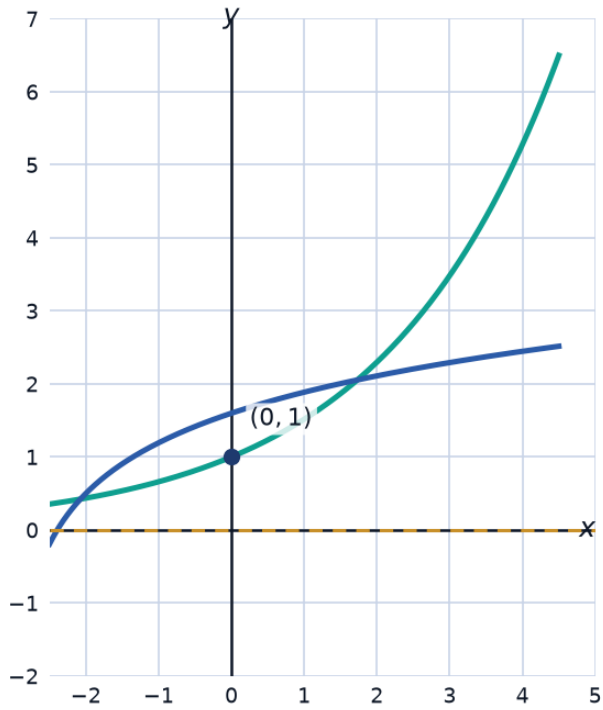
15 [Challenging] • Target: Continuity on an interval • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Find the directional derivative of $f(x, y) = x\cos(y) - y$ at $(-2, \ -2)$ in the direction of $\langle -3, 4 \rangle$.

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16 [Challenging] • Target: Applying the Intermediate Value Theorem • Cross-Lesson: 11.01 Sequences and Series

If $z = \ln(x^2 + y^2)$ with $x = t^2$ and $y = \sin(t)$, find $\frac{dz}{dt}$ at $t = 1$.



17 [Challenging] • Target: Visualizing discontinuities • Cross-Lesson: 12.01 Convergence Tests

Evaluate $\int_0^{\pi} e^{3x}\cos(2x)\,dx$.

18 [Challenging] • Target: Continuity of composite and combined functions • Cross-Lesson: 12.02 Power Series and Taylor Series

Flux of $\mathbf{F} = (x, y, 0)$ through lateral surface of cylinder $x^2 + y^2 = 1$, $0 \le z \le 2$, outward:

19 [Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Evaluate $\int_0^{\pi} e^{3x}\sin(2x)\,dx$.

20 [Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Evaluate $\int_0^{\pi} e^{-x}\cos(2x)\,dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Definition of Continuity at a Point

Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ is defined, $\lim_{x \to c} f(x)$ exists, and $\lim_{x \to c} f(x) = f(c)$

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

d. $f(c) = 0$ and $f'(c) = 0$

22 [Foundational] • Target: Definition of Continuity at a Point

Determine if $f(x)$, defined as $\frac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $f(2) = 5$, is continuous at $x = 2$.

a. Discontinuous (removable discontinuity at $x = 2$)

b. Continuous at $x = 2$

c. Discontinuous (jump discontinuity)

d. Discontinuous (infinite discontinuity)

23 [Foundational] • Target: Definition of Continuity at a Point

For what value of a is $h(x)$, defined as $ax^2 + 1$ for $x \leq 2$ and $3x - a$ for $x > 2$, continuous everywhere?

a. 1

b. 2

c. 3/5

d. -1

24 [Foundational] • Target: Classification of Discontinuities

Classify the discontinuity of $g(x) = \frac{x - 1}{|x - 1|}$ at $x = 1$.

a. Jump discontinuity

b. Removable discontinuity

c. Infinite discontinuity

d. Essential oscillating discontinuity

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25 [Foundational] • Target: Classification of Discontinuities

Classify the discontinuity of $f(x) = \frac{1}{(x-4)^2}$ at $x = 4$.

a. Infinite discontinuity

b. Removable discontinuity

c. Jump discontinuity

d. Continuous

26 [Foundational] • Target: Classification of Discontinuities

How can $f(x) = \frac{\sin(3x)}{x}$ be defined at $x = 0$ to make it continuous?

a. Set $f(0) = 3$

b. Set $f(0) = 1$

c. Set $f(0) = 0$

d. It cannot be made continuous

27 [Foundational] • Target: Intermediate Value Theorem (IVT)

State the primary hypothesis required to apply the Intermediate Value Theorem to f on $[a, b]$.

a. f must be continuous on the closed interval $[a, b]$

b. f must be differentiable on (a, b)

c. $f(a) = f(b)$

d. $f'(x) > 0$ on $[a, b]$

28 [Foundational] • Target: Intermediate Value Theorem (IVT)

Use the IVT to show that $x^3 + 2x - 5 = 0$ has a root in which interval?

a. $(1, 2)$

b. $(0, 1)$

c. $(2, 3)$

d. $(-1, 0)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Intermediate Value Theorem (IVT)

A continuous function g satisfies $g(0) = -3$ and $g(4) = 5$. Does $g(x) = \pi$ have a solution on $(0, 4)$?

a. Yes, because $-3 < \pi < 5$ and g is continuous

b. No, because π is irrational

c. Only if g is differentiable

d. Cannot be determined without $g'(x)$

30 [Intermediate] • Target: Extreme Value Theorem (EVT) and Properties

What does the Extreme Value Theorem guarantee for a function continuous on a closed bounded interval $[a, b]$?

a. f attains both an absolute maximum and an absolute minimum on $[a, b]$

b. f has at least one inflection point

c. f is invertible on $[a, b]$

d. $f'(c) = 0$ for some $c \in (a, b)$

31 [Intermediate] • Target: Extreme Value Theorem (EVT) and Properties

Which function fails to achieve an absolute maximum on $(0, 1)$?

a. $f(x) = \frac{1}{x}$

b. $f(x) = x^2$

c. Both $f(x) = 1/x$ and $f(x) = x^2$

d. $f(x) = 5$

32 [Intermediate] • Target: Extreme Value Theorem (EVT) and Properties

If f and g are continuous at $x = c$, which of the following is NOT guaranteed to be continuous at $x = c$?

a. $f(x)/g(x)$

b. $f(x) + g(x)$

c. $f(x) \cdot g(x)$

d. $[f(x)]^2$

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33 [Intermediate] • Target: Testing continuity

A continuous function on a closed, bounded interval $[a, b]$ is guaranteed (Extreme Value Theorem) to be:

a. bounded and to attain its max and min

b. monotonic

c. unbounded

d. differentiable

34 [Intermediate] • Target: Types of discontinuity and the Intermediate Value Theorem

Removable discontinuity: $f(x) = \frac{x^2-4}{x-2}$. To make f continuous at $x = 2$, define $f(2) =$

a. 4

b. 2

c. 0

d. -4

35 [Intermediate] • Target: Continuity on an interval

By the Intermediate Value Theorem, does $f(x) = x^3 - 4x - 2$ have a root in $[2, 3]$?

a. No — $f(2)$ and $f(3)$ have the same sign

b. Yes — $f(2)$ and $f(3)$ have opposite signs

c. No — f is not continuous on $[2, 3]$

d. Cannot be determined without more information

36 [Intermediate] • Target: Applying the Intermediate Value Theorem

By the Intermediate Value Theorem, does $f(x) = x^5 - x - 1$ have a root in $[1, 2]$?

a. No — $f(1)$ and $f(2)$ have the same sign

b. Cannot be determined without more information

c. Yes — $f(1)$ and $f(2)$ have opposite signs

d. No — f is not continuous on $[1, 2]$

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37 [Intermediate] • Target: Visualizing discontinuities

By the Intermediate Value Theorem, does $f(x) = x^4 - x - 1$ have a root in $[1, 2]$?

a. Yes — $f(1)$ and $f(2)$ have opposite signs

b. No — $f(1)$ and $f(2)$ have the same sign

c. Cannot be determined without more information

d. No — f is not continuous on $[1, 2]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Continuity of composite and combined functions • Cross-Lesson: 12.02 Power Series and Taylor Series

By the Intermediate Value Theorem, does $f(x) = x^3 - 2x - 5$ have a root in $[2, 3]$?

a. Yes — $f(2)$ and $f(3)$ have opposite signs

b. Cannot be determined without more information

c. No — f is not continuous on $[2, 3]$

d. No — $f(2)$ and $f(3)$ have the same sign

39 [Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

By the Intermediate Value Theorem, does $f(x) = x^3 + 4x^2 - 10$ have a root in $[1, 2]$?

a. No — f is not continuous on $[1, 2]$

b. Cannot be determined without more information

c. Yes — $f(1)$ and $f(2)$ have opposite signs

d. No — $f(1)$ and $f(2)$ have the same sign

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40 [Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

By the Intermediate Value Theorem, does $f(x) = x^3 - 6x + 2$ have a root in $[2, 3]$?

a. Cannot be determined without more information

b. No — f is not continuous on $[2, 3]$

c. Yes — $f(2)$ and $f(3)$ have opposite signs

d. No — $f(2)$ and $f(3)$ have the same sign

41 [Challenging] • Target: Definition of Continuity at a Point • Cross-Lesson: 15.01 Double Integrals

By the Intermediate Value Theorem, does $f(x) = x^5 - 3x - 1$ have a root in $[1, 2]$?

a. No — $f(1)$ and $f(2)$ have the same sign

b. No — f is not continuous on $[1, 2]$

c. Cannot be determined without more information

d. Yes — $f(1)$ and $f(2)$ have opposite signs

42 [Challenging] • Target: Classification of Discontinuities • Cross-Lesson: 16.01 Vector Fields and Line Integrals

By the Intermediate Value Theorem, does $f(x) = x^3 - 5x - 2$ have a root in $[2, 3]$?

a. Cannot be determined without more information

b. No — $f(2)$ and $f(3)$ have the same sign

c. No — f is not continuous on $[2, 3]$

d. Yes — $f(2)$ and $f(3)$ have opposite signs

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43 [Challenging] • Target: Intermediate Value Theorem (IVT) • Cross-Lesson: 2.01 Evaluating Limits

By the Intermediate Value Theorem, does $f(x) = x^5 - 4x - 1$ have a root in $[1, 2]$?

a. No — f is not continuous on $[1, 2]$

b. Cannot be determined without more information

c. Yes — $f(1)$ and $f(2)$ have opposite signs

d. No — $f(1)$ and $f(2)$ have the same sign

44 [Challenging] • Target: Extreme Value Theorem (EVT) and Properties • Cross-Lesson: 3.01 The Derivative from First Principles

By the Intermediate Value Theorem, does $f(x) = x^5 - 2x - 3$ have a root in $[1, 2]$?

a. Cannot be determined without more information

b. No — $f(1)$ and $f(2)$ have the same sign

c. No — f is not continuous on $[1, 2]$

d. Yes — $f(1)$ and $f(2)$ have opposite signs

45 [Challenging] • Target: Testing continuity • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

By the Intermediate Value Theorem, does $f(x) = x^3 + 4x - 15$ have a root in $[1, 2]$?

a. Cannot be determined without more information

b. No — $f(1)$ and $f(2)$ have the same sign

c. No — f is not continuous on $[1, 2]$

d. Yes — $f(1)$ and $f(2)$ have opposite signs

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